

Module 04 - Shareable Invite Links

Secure join links with expiry and revocation

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Allow group admins/owners to create shareable links, copy them, revoke them, and allow authenticated users to join the group through `/join/:inviteCode`.

Invite document

```
inviteLinks/{inviteId}
{
  groupId,
  createdBy,
  codeHash,
  createdAt,
  expiresAt,
  maxUses,
  usesCount,
  revokedAt
}
```

Security rule

Do not store a raw secret invite code if it is not necessary. Prefer a non-guessable token, or store a hash and validate the presented token through trusted backend logic when stronger protection is needed.

Flow

```
Admin clicks Invite
-> Create token
-> Copy /join/{token}
-> Recipient opens link
-> Authenticate if needed
-> Validate invite
-> Confirm group details
-> Join
-> Create membership
-> Increment usage
-> Write activity
```

Edge cases

- Expired link
- Revoked link
- Max uses reached
- Already a member
- Nonexistent token
- Unauthenticated visitor
- User tries to reuse a one-time link

Acceptance criteria

- Generated link opens a join preview.
- Valid authenticated user joins once.
- Expired/revoked links cannot join.
- Member list and activity feed update after join.
- Admin can revoke a link.
- Link creation itself is permission-protected.